PAVAN KUMAR NAYAKANTI

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PROFESSIONAL SUMMARY

Cloud architect and platform engineering leader with 20+ years designing multi-cloud reference architectures, HA/DR patterns, resiliency frameworks, and compliance guardrails across regulated FinTech and banking environments (AWS, Azure, GCP). Author of the Cloud Platform Maturity Index (CPMI) -- an enterprise cloud architecture framework defining capability standards and guardrails deployed across three major banking clients. Proven track record translating architectural intent into operational reality: HA/DR designed upfront into CDK-driven AWS migrations, resiliency validated through Chaos Engineering (Keptn), and compliance encoded as CI/CD guardrails (SOX/GDPR). Strong multi-cloud certifications (AZ-305, SAA-C01, GCP PCA) and 20+ years of hands-on production experience backing every architectural recommendation.

CORE COMPETENCIES

Multi-Cloud Architecture (AWS / Azure / GCP) | Cloud Platform Maturity Index (CPMI) -- Framework Author | HA/DR & Availability Pattern Design | Resiliency Engineering & Chaos Engineering (Keptn) | Landing Zone Design & Cloud Guardrails | IaC Standards (Terraform, AWS CDK) | Compliance-as-Code (SOX, GDPR) | Architecture Governance & Design Standards | SLO/SLA Definition & Dependency Modeling | DevSecOps & Security Architecture | SRE -- Observability, On-Call, Postmortems | Executive & CTO-Level Stakeholder Influence

WORK EXPERIENCE

IRIS Software

Sep 2020 -- Present

Sr. Specialist -- Cloud Practice & SRE Lead | India (Clients: ADP, Charles Schwab, BMO)

- **Cloud Platform Maturity Index (CPMI) -- Architecture Framework:** Designed and deployed the CPMI -- an enterprise cloud architecture framework defining capability standards, compliance guardrails, and maturity thresholds across three major banking clients (ADP, Charles Schwab, BMO). Drove measurable improvement in cloud security, efficiency, and capability maturity organisation-wide.
- **HA/DR Architecture -- Designed Upfront (BMO):** Designed HA/DR architecture for BMO CCB cloud migration from day one -- multi-AZ ECS Fargate and Lambda deployment via AWS CDK with automated failover, Disaster Recovery objectives defined upfront and tested during migration. Led end-to-end on-premise to AWS migration with no production incidents during cutover.
- **Resiliency Validation via Chaos Engineering (ADP):** Established a Chaos Engineering programme (Keptn) as the architectural validation layer for ADP's SRE platform -- fault injection experiments across network, compute, and dependency tiers; documented failure modes and hardened resiliency patterns before production traffic. Proactively identified fault scenarios that informed platform redesigns.
- **SLO/SLA Definition & Observability Architecture (ADP):** Defined SLOs, SLIs, and error budgets across multi-client cloud platforms; deployed Prometheus, Grafana, Dynatrace, and MS Application Insights for full-stack observability with AlertManager-driven on-call pipelines. Scaled ADP's SRE organisation from 4 to 40 engineers; reduced MTTR 40%+ through AIOps-driven automated remediation.
- **GCP Landing Zone Architecture (Charles Schwab):** Led GCP landing zone preparation and workload migration POCs for Charles Schwab -- evaluated GCP-native services for cloud-native deployments; executed cross-platform artifact deployment alignment across TD Ameritrade and Charles Schwab environments as part of the merger integration programme.
- **FinOps & Cloud Cost Architecture (BMO):** Drove AWS FinOps across BMO estate -- rightsized workloads to match actual demand, identified unused resources, reduced cloud waste. GenAI risk assistants deployed on AWS Bedrock via event-driven pipelines.
- **DevSecOps as a Service:** Architected enterprise-grade DevOps as a Service platform -- CI/CD standards, IaC blueprints, observability patterns, and security guardrails delivered as reusable reference implementations across clients.

IRIS Software

Jun 2018 -- Sep 2020

Automation Technology Specialist | Toronto, Canada (Client: Royal Bank of Canada Insurance)

- **Compliance Architecture -- Guardrails in CI/CD (SOX/GDPR):** Translated SOX and GDPR compliance requirements into enforceable CI/CD guardrails across 250+ RBC applications -- SAST/DAST/IAST gates (SonarQube, JFrog), approval workflows, and audit trail pipelines baked into Jenkins/UrbanCode/GitHub. Zero manual compliance workarounds. Application onboarding reduced from 5 weeks to 5 hours (98% reduction).
- **DORA Framework & Architecture Standards:** Established DORA as the engineering health standard across 60+ RBC teams via Azure DevOps -- Deployment Frequency, Lead Time for Changes, Change Failure Rate, and MTTR as shared architecture metrics. Measurable improvement in delivery performance reviewed quarterly with VP stakeholders.
- Established SRE principles and observability architecture standards within RBC's DevSecOps programme -- incident classification frameworks, reliability-focused deployment gates, and Splunk log aggregation pipelines across 250+ applications.
- Led enterprise migration of 250+ applications across 60+ teams from CA SCM/Harvest to GitHub; contributed to ~$200K in new project acquisitions through pre-sales and architecture leadership.

IRIS Software

Feb 2017 -- Jun 2018

Automation Architect | Toronto, Canada (Clients: Insurance & Capital Markets)

- Designed fault-tolerant, container-orchestrated deployment architectures using Kubernetes and Helm -- zero-downtime rollout patterns with RBAC, resource governance, and dependency isolation for capital markets platforms.
- Led RPA architecture and practice delivery -- 80+ applications, 250+ components across heterogeneous stacks. Contributed to $478K SOW closure and $180K Capital Markets engagement.

TechMahindra Pvt. Ltd.

Jul 2015 -- Feb 2017

Technology Manager | Noida, India (Client: Adobe)

- Architected and delivered zero-downtime data centre relocation for Adobe -- full migration plan, dependency mapping, cutover execution, and business continuity across all production workloads with no service disruption.
- Led SAP HANA upgrades, Informatica-to-SQL Server data migration (schema redesign, ETL re-engineering, enterprise-scale data validation), and platform modernisation.

Microsoft Corporation

Jul 2007 -- Dec 2014

Sr. SDET -- QA, Automation & Production Engineering | India

- Full production ownership of Sales & Marketing platforms (US, EMEA, APAC) -- 99.99% uptime over 7 years. Owned all non-functional architecture objectives: performance, availability, scalability, security across network, infrastructure, and application tiers. Supported 200% user growth through self-healing architectures and cloud migration with no reliability degradation.

KEY ARCHITECTURE INITIATIVES

Cloud Platform Maturity Index (CPMI) -- Enterprise Architecture Framework

Reference Architecture -- AWS / Azure / GCP -- 3 Clients

Designed the CPMI -- an enterprise cloud architecture framework defining capability standards, compliance guardrails, and maturity thresholds. Deployed across ADP, Charles Schwab, and BMO; drove measurable cloud security, efficiency, and capability maturity improvements. Encoded as reusable reference implementations for adoption without mandate.

BMO AWS Migration -- HA/DR Designed Upfront

AWS CDK -- ECS Fargate -- Lambda -- Multi-AZ -- DR

Designed HA/DR architecture from day one for BMO CCB cloud migration -- multi-AZ ECS Fargate and Lambda via CDK with automated failover and defined recovery objectives. Full on-premise-to-AWS migration delivered with zero production incidents during cutover.

Resiliency Architecture Validation -- Chaos Engineering (ADP)

Keptn -- Fault Injection -- Dependency Modeling -- SRE

Established Chaos Engineering (Keptn) as the validation layer for ADP's platform resiliency architecture. Fault injection experiments across network, compute, and dependency tiers; documented failure modes and remediation patterns; hardened architectures before production traffic. MTTR reduced 40%+.

Compliance Guardrails in CI/CD -- RBC (250+ Apps, SOX/GDPR)

Compliance-as-Code -- DevSecOps -- Azure DevOps

Translated SOX and GDPR requirements into enforceable pipeline guardrails across 250+ RBC applications -- SAST/DAST/IAST scanning, approval workflows, and audit trail automation. Zero manual workarounds. Application onboarding reduced from 5 weeks to 5 hours.

EDUCATION

Masters in Artificial Intelligence -- Woolf University 2024

General Management Program (GMP) -- ISB Hyderabad 2022

M.Sc in Computer Science -- Osmania University 2002

B.Sc in Electronics -- Kakatiya University 2000

CERTIFICATIONS

Microsoft Certified: Azure Solutions Architect Expert (AZ-305) 2026

AWS Certified Solutions Architect -- Associate (SAA-C01) 2026

Google Cloud Certified -- Professional Cloud Architect (PCA) 2026

Certified Kubernetes Application Developer (CKAD) 2025

Microsoft Certified: Azure AI Engineer Associate (AI-102)

ITIL® Certificate in IT Service Operations

PMI Agile Certified Professional (PMI-ACP)

TECHNICAL SKILLS

Cloud Architecture: AWS (AZ-305), Azure (AZ-305), GCP (PCA) -- multi-cloud reference architectures, landing zone design, PaaS patterns, multi-region HA/DR, cloud guardrails; IaC standards (Terraform, AWS CDK, Helm)

Resiliency & Availability: SLI/SLO/SLA definition, Error Budgets, Chaos Engineering (Keptn), self-healing frameworks, dependency modeling, fault injection validation, MTTR governance

Compliance & Security Architecture: Compliance-as-code (SOX, GDPR); DevSecOps guardrails (SAST/DAST/IAST, SonarQube, JFrog); ITIL-aligned change governance; audit trail automation; network segmentation and least-privilege IAM (Azure, AWS)

Observability & Platform Standards: Prometheus + AlertManager, Grafana, Dynatrace, MS Application Insights / Azure Log Analytics, Splunk; DORA metrics; architecture governance frameworks (CPMI)

Container & Delivery Platform: Kubernetes (CKAD), Docker, Helm, Azure DevOps, GitHub Actions, Jenkins; CI/CD standards and reusable pipeline templates

AI & Automation: GenAI (AWS Bedrock, Azure AI), AIOps; Python, Bash, PowerShell, Go; MSc in Artificial Intelligence